
Enhancing Degraded Scene Text Recognition through Text Prior Guided Super-Resolution

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1 Introduction

Scene Text Recognition (STR) is the process of recognizing text sequences in scene images. It has a practical role in bridging the gap between computer vision and natural language processing for systems to interpret and process textual information in real-life environments. STR is harder and requires more complexity than generic image classification, because it requires processing sequential text information and context.

Standard STR models achieve high accuracy on clear, high-quality text datasets, but degradations that obscure characters and disrupt sequence modeling are detrimental to performance. These main degradations include low resolution from lower quality hardware, blurring from focus issues or object motion, random noise from image sensors, and compression artifacts from JPEG compression. These types of degradations in severe intensities can cause serious detrimental effects to text visibility, structure, and sequencing that can break or diminish recognition processing from standard models or pipelines.

Our primary objective is to simulate, quantify, analyze, and mitigate the text recognition accuracy performance loss in STR models caused by these real-world heavy degradations in scene images. Many models are benchmarked on largely clean datasets, and degraded inputs are less studied, so we intend to bridge this research gap by simulating and comparing the effectiveness of STR models with clean, degraded, and restored image inputs using synthetic degradation, an irregular text dataset, and the Text Prior Guided Super-Resolution module (TPGSR).

Unlike generic super-resolution methods that are intended for the general sharpening of a lower resolution image, TPGSR utilizes text priors from a recognition model to guide the restoration phase more optimally for STR. The TPGSR module is intended to explore the degradation resiliency aspect of this pipeline and how text-guided restoration of heavily degraded inputs affects different recognition strengths. Further exploration of this topic can have real-world applications, such as autonomous vehicles or aircraft reading signs in inclement weather.

Open source code disclosure:

SVTRv2 model recognition pipeline, configs, and pretrained weights:
<https://github.com/Topdu/OpenOCR>

CRNN and TRBA benchmark model pretrained weights:
<https://github.com/clovaai/deep-text-recognition-benchmark>

36 2 Literature Survey

37 Long et al. [1] provide a comprehensive survey on scene text detection and recognition, covering the
38 evolution from traditional methods to deep learning approaches and establishing standard benchmarks
39 and evaluation protocols. Liu et al. [2] complement this with a focused review on CNN and
40 RNN-based architectures, categorizing methods into detection, recognition, and end-to-end systems.
41 On the methodology side, Du et al. [3] present SVTR, a single visual model that eliminates sequential
42 modeling through patch-wise tokenization and hierarchical mixing, achieving state-of-the-art accuracy
43 with better efficiency.
44 Wang et al. [4] introduce TextZoom, the first dataset of paired low-resolution and high-resolution
45 scene text images captured at real camera zoom levels, enabling principled evaluation of text super-
46 resolution methods. The dataset also provides strong baselines for scene text image super-resolution.
47 Ma et al. [5] propose TPGSR (Text Prior Guided Super-Resolution), which conditions a super-
48 resolution network on features from a frozen text recognizer. Their key insight is that generic
49 super-resolution loss functions (such as L1 or MSE) produce pixel-accurate but not necessarily
50 readable outputs; by adding a recognition loss computed through a frozen recognition model, the
51 SR output is encouraged to preserve text semantics. Their evaluation demonstrates significant
52 improvements in recognition accuracy on challenging TextZoom subsets. Shi et al. [6] propose
53 CRNN, a pioneering end-to-end scene text recognition framework that integrates convolutional
54 feature extraction with recurrent sequence modeling and CTC-based transcription. This architecture
55 eliminates the need for character-level segmentation and serves as a standard baseline for evaluating
56 modern recognition systems and text-aware super-resolution methods.

57 3 Methods

58 3.1 Overview

59 Our pipeline has four trainable components that are combined at inference time. A Text Prior
60 Generator (TPG) is a recognizer trained on clean text and then frozen; its role is to serve as a
61 "judge" during super-resolution training, providing a recognition-based supervision signal. A Text
62 Super-Resolution Network (TSRN) is a compact convolutional network with recurrent residual blocks
63 that performs $2\times$ upscaling from 16×64 low-resolution input to 32×128 high-resolution output. A
64 baseline recognizer is trained on heavily degraded crops directly, without super-resolution, serving
65 as the reference against which TPGSR is measured. A fine-tuned recognizer is the baseline further
66 adapted on super-resolved outputs produced by TSRN.

67 The method operates in four training phases and one inference phase. Phase 0 trains the TPG on clean
68 crops. Phase 1 trains TSRN with frozen TPG guidance using a combined loss of L1, MSE, and cross-
69 entropy (we use L1 and MSE for pixel losses and cross-entropy accounts for text recognizer losses).
70 Phase 2 runs TSRN on the full dataset to cache super-resolved outputs, with a width-filtered hybrid
71 pass-through that skips crops outside the $[40, 160]$ pixel width range. This filter is needed because
72 images below this width don't scale properly and images above this width break our model (dimension
73 mismatch). The optimal range has been determined based on experiments. Phase 3 fine-tunes the
74 baseline recognizer on this SR cache. Phase 4 evaluates the fine-tuned recognizer on test SR outputs
75 and compares against the baseline recognizer on raw degraded test inputs.

76 3.2 Intuition: why text-guided restoration over generic SR

77 Generic super-resolution methods minimize pixel-level losses, typically L1 or MSE between the
78 SR output and a high-resolution ground truth. For natural images with rich texture, this objective
79 produces visually satisfactory results. For text, it fails in a specific way: when training data contains
80 character-level ambiguity (example a given pixel pattern that looks like "O" can plausibly be recognized
81 as "0").

82
 83 TPGSR addresses this by adding a recognition loss term(generated by our "judge"): the SR
 84 output is passed through a frozen text recognizer(TPG), and cross-entropy is computed against the
 85 ground truth text. This forces TSRN to produce outputs that are not just pixel-accurate but also
 86 unambiguously readable. The frozen recognizer is **not trained** during TSRN training.

87 3.3 Data preparation

88 We use the Total-Text and CTW1500 datasets, both chosen for their irregular text characteristics
 89 including curved words, non-horizontal orientations, and varied fonts that reflect real-world scene
 90 text better than rectangular benchmark crops. From the annotated polygon coordinates, we extract
 91 word-level tight crops. Each sample is stored with three associated fields: the cropped RGB image,
 92 the ground-truth text string, and a binary polygon mask indicating which pixels within the bounding
 93 box contain text versus background.

94
 95 The polygon mask is **critical** to our pipeline. Unlike TextZoom dataset(dataset used in
 96 TPGSR paper) where the rectangular boxes contains text with minimal background noise, our
 97 data is more realistic and scene text often occupies only a fraction of its axis-aligned bounding
 98 rectangle (like a curved text leave triangular background regions in the corners, and rotated text
 99 leaves even larger unused areas). Running SR directly on such crops would waste the model’s
 100 capacity on background pixels, that could better be spent sharpening the text features. We ad-
 101 dress this problem by feeding the mask as **fourth input channel** along the RGB channels to the TSRN.

102
 103 For each clean crop, we synthesize a corresponding heavily degraded version using a com-
 104 pound degradation pipeline. The pipeline samples 2-3 effects per crop from a set of five: JPEG
 105 compression (quality uniformly sampled from [28, 52]), additive Gaussian noise (standard deviation
 106 from [16, 30]), downsample-upsample (factor from [1.8, 2.8]), Gaussian blur (kernel size from [3,
 107 7]), and motion blur (kernel size from [3, 7]). Compounding multiple effects per sample creates a
 108 harder and more diverse degradation distribution than either single-effect synthesis or TextZoom’s
 109 camera-zoom-only degradation. After preprocessing, our dataset contains approximately 88,000
 110 training crops, 11,000 validation crops, and 10,000 test crops, each with clean, degraded, and mask
 111 variants.

112 3.4 TSRN architecture

113 TSRN is a relatively compact network with approximately 780K parameters. The input is
 114 a 4-channel tensor at 16×64 resolution (3 RGB channels concatenated with the 1-channel
 115 polygon mask). A head convolution with a 9×9 kernel embeds this into a 64-channel feature
 116 map. An optional Spatial Transformer Network predicts affine parameters that rectify geomet-
 117 ric distortion, initialized to identity so it begins as a no-op and learns small corrections during training.

118
 119 The feature map then passes through five recurrent residual blocks. Each block contains
 120 two 3×3 convolutions with batch normalization and a PReLU nonlinearity, with a skip connection,
 121 followed by a bidirectional GRU that sweeps horizontally across width. The BiGRU is the key
 122 architectural choice that distinguishes TSRN from generic SR networks: text is horizontal and
 123 sequential, so features at each spatial column depend on neighboring columns for stroke continuity.
 124 The BiGRU provides this inductive bias, which a standard convolutional network lacks.

125
 126 When in iteration 2 or later (see training procedure below in 3.5), a text prior feature map
 127 from the TPG is fused into TSRN’s intermediate features via channel-wise concatenation followed by
 128 a 3×3 projection convolution. The fused features continue through the recurrent residual blocks.

129
 130 Upsampling is performed with a factor of 2: a convolution projects 64 channels to 256,
 131 then we rearrange the extra 4 channels into a 2×2 spatial block, producing 64 channels at 32×128. A

final tail convolution with a 9×9 kernel produces the 3-channel RGB output, with Tanh activation bounding it to [-1, 1].

3.5 TPGSR training procedure

TPG (Phase 0) is trained before the TPGSR pipeline begins. For training TSRN, there are two passes that TSRN runs Iteration 1 and Iteration 2.

Iteration 1: In this pass the TSRN produces an initial **super-resolved** output SR. Note: this stage doesn't have any text prior yet. This produces a pixel-accurate image but doesn't have text-awareness at this stage. This output is passed to our frozen TPG(Text Prior Generator). This frozen TPG produces attention logits(for TRBA) or CTC log-probabilites(for CRNN) describing what characters it believes are present in the SR output. These logits are projected via a small linear network to a 32-channel spatial feature map, tiled to match TSRN's 16×64 feature grid.

Iteration 2: In the second iteration TSRN receives the same LR + mask input but this time it has the text prior fused in its intermediate features. This produces the refined SR output. The TPG is then again run a second time on our refined SR output, but this time we have teacher forcing in play. The ground truth is fed in as the decoder's previous-token inputs at each timestep. The resulting character logits are used to compute the recognition loss.

Teacher forcing is used during training for two reasons. First, it prevents error cascading: if TPG's attention decoder made a prediction error at timestep 1 and fed it to timestep 2 autoregressively, the entire sequence would drift, producing noisy gradients that would reflect TPG's own decoding errors rather than genuine SR quality issues. Second, it produces clean **per-timestep supervision**: every position receives a direct gradient signal from ground truth, maximizing the information density of the recognition loss.

The total loss is a weighted sum:

- LR : Low-resolution input image
- HR : High-resolution ground-truth image
- SR_2 : Final super-resolved output produced by TSRN (Iteration 2)
- SR : Intermediate super-resolved output produced in Iteration 1
- $TPG(\cdot)$: Frozen Text Prior Generator network
- y_{tf} : Teacher-forcing input sequence used for autoregressive decoding
- y_{target} : Ground-truth text label sequence

$$L = \lambda_{L1} \cdot L_{L1}(SR_2, HR) + \lambda_{MSE} \cdot L_{MSE}(SR_2, HR) + \lambda_{CE} \cdot L_{CE}(TPG(SR_2, y_{tf}), y_{target})$$

$$\lambda_{L1} = \lambda_{MSE} = 1.0, \quad \lambda_{CE} = 0.1.$$

The lower CE weight prevents the recognition objective from overwhelming pixel reconstruction. Without it, TSRN could produce outputs that are readable by the recognizer but visually unlike HR. Gradients flow **only** to TSRN (TPG parameters are frozen throughout).

3.6 Hybrid inference with width filtering

During Phase 2 (SR caching) and at test time, we apply a hybrid strategy. Crops whose original width falls inside [40, 160] pixels are passed through TSRN to produce SR outputs. Crops outside this range are "passed through" to the downstream recognizer without super-resolution.

This filtering reflects a geometric constraint. The 40-pixel lower bound corresponds to the

point where downsampling to TSRN’s 16×64 input leaves fewer than 2 pixels per character, below which recovery is not feasible. The 160-pixel upper bound corresponds to the point where fitting into TSRN’s 128-pixel output requires aggressive aspect-ratio compression that destroys character readability. On our dataset, approximately 43% of crops fall inside this range and receive SR treatment, while 57% pass through. Unlike TextZoom, where all crops are pre-standardized to fixed dimensions, real scene text has a long-tailed width distribution that necessitates this routing decision.

3.7 Cross-architecture evaluation

We train the full TPGR pipeline separately for CRNN and TRBA. Each uses its own TPG(matching the recognizer architecture) and their respective TSRN instance trained with that TPG as a "judge". This principle of matched TPG-recognizer ensures the super-resolution is optimized for the specific feature preferences of each downstream model. A CRNN trained on SR outputs guided by a TRBA-TPG would see inputs optimized for attention-based reading patterns that its CTC-based pipeline doesn’t leverage.

For SVTRv2, we evaluate a **hybrid** configuration: the TRBA-guided TSRN is reused to produce SR outputs, which are then used to fine-tune SVTRv2. This does not match the "ideal" matched-pair design but tests whether SR outputs generalize across recognizer families.

4 Experiments

For our experiments, the main metric that we judged performance by was text recognition accuracy on the test set. From preprocessing and restoration, we prepared the clean inputs, degraded inputs, and restored inputs separately for STR recognition training and inference, and compared the results to each other after hyperparameter tuning.

Alongside the main question of how much restoration affects STR recognition accuracy, we sought to determine how much degradation affects STR recognition accuracy and how significant degradation and restoration are based on the sophistication of the base models. We also wanted to subjectively see how visually effective the restoration process was after generating sample crops and comparing the clean, degraded, and restored versions side by side.

Below are several sample word crops with all of their respective clean, degraded, and restored versions for comparison:



Figure 1: Sample Image of Word Crop of Text "brewed"



Figure 2: Sample Image of Word Crop of Text "SAWMILL"

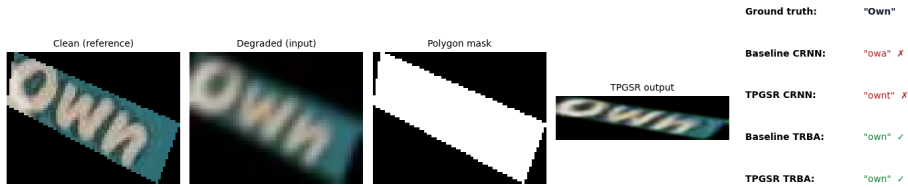


Figure 3: Sample Image of Word Crop of Text "Own"

From these examples, we can infer that while TPGSR restoration displays visible and moderate gains in restoring degraded crops, it is not perfect and there is still some slight residual degradation present.

Our experimental setup includes training and inference of all 3 models of CRNN, TRBA, and SVTRv2 on:

1. Clean Inputs for reference generic STR
2. Degraded Inputs as a baseline
3. Restored Inputs after TPGSR fine-tuning

The main hyperparameters for which we saw defined improvement for these models were fine-tuned based on all 3 input types and are as follows:

1. Epochs
2. Batch Size
3. Learning Rate
4. Weight Decay

Model	Epochs	Batch Size	Learning Rate	Weight Decay
CRNN	12	64	$2e^{-4}$	$5e^{-5}$
TRBA	5	64	$2e^{-4}$	$1e^{-4}$
SVTRv2	5	8	$1e^{-5}$	$1e^{-5}$

These hyperparameters for each model we fine-tuned were what we found to perform the best across all 3 types of inputs. One of the main takeaways is that CRNN gained the most from a longer training schedule than the other two models. CRNN appeared to gain the most from a longer training schedule, because it is the simplest of the three recognizers and relies on gradually learning features through its CNN and BiLSTM pipeline. The other main takeaway was that SVTRv2

benefited most from a smaller batch size, because its fine-tuning set is relatively small compared to its large modeling capacity as a fully transformer-based architecture. This helped generalization to the validation and test set, reaching higher accuracies.

Model	Clean Inputs	Degraded Inputs	Restored Inputs
CRNN	57.30%	34.30%	39.60%
TRBA	65.00%	42.10%	44.10%
SVTRv2	82.90%	57.30%	59.00%

From our results, we can see clear trends in terms of the model performances across all inputs, degradation effects, and restoration effects. All three models perform best on clean inputs, which is expected because STR models are generally optimized for clean and high resolution images. Once heavy degradations are introduced, accuracies across all 3 models suffer substantial dropoffs of 22% – 26%, showing how significantly detrimental our synthetic degradations are to base scene text recognition.

SVTRv2 performed the best on all 3 inputs, followed by TRBA, and then CRNN. This suggests that newer and more sophisticated, semantically guided, transformer architectures have stronger feature extraction and sequence modeling capacity than the older baselines. This performance and reasoning stays consistent under heavily degraded inputs.

The restoration results show that TPGSR provides a modest benefit to all three models. Compared with degraded inputs, restored inputs increase CRNN accuracy by 5.3%, TRBA accuracy by 2.0%, and SVTRv2 accuracy by 1.7%. These improvements indicate that TPGSR is able to recover some useful textual structure that is lost under heavy degradation, although restoration does not return performance to clean-input levels. The improvement also differs by model, with CRNN gaining the most, TRBA gaining moderately less, and SVTRv2 gaining slightly even less. This could be because weaker recognizers inherently perform worse against degraded, low resolution images and are less degradation resilient than newer, more sophisticated recognizers, so they have the most to gain from a restoration stage, even if residual degradation remains.

Overall, the table shows three main conclusions: heavy degradation substantially harms scene text recognition, stronger recognizer architectures remain more resilient under corruption, and TPGSR consistently improves recognition accuracy across all tested models. These findings support the claim that text-guided restoration is a useful stage for degraded STR, even though it does not fully eliminate the performance gap between restored and clean images.

5 Conclusion and Future Work

We evaluated Text Prior Guided Super-Resolution across three recognizer architectures (CRNN, TRBA and SVTR) of increasing strength (baseline accuracies), testing whether SR preprocessing helps recover accuracy lost to heavy synthetic degradation on real scene text with polygon-annotated irregular geometry. Our results confirm that TPGSR improves accuracy across all tested architectures, but reveal an **inverse scaling relationship**: the simpler the baseline recognizer, the larger the gain from TPGSR. The strongest recognizer with SR preprocessing still achieves the highest absolute accuracy, but receives the smallest relative benefit from the preprocessing step.

Two directions for future work seem most promising. First, training a dedicated SVTRv2-TPG by integrating our TPGSR pipeline with OpenOCR’s training infrastructure would test whether the inverse scaling pattern persists under matched-pair evaluation, or whether it partly reflects the mismatch in our current SVTRv2 hybrid. Second, evaluating on genuinely real-world degraded captures (not synthetic degradation) would confirm whether the observed patterns transfer to deployment settings.

Beyond these extensions, our cross-architecture analysis suggests a broader pattern worth further study: as recognition models continue to grow stronger (like SVTR), preprocessing techniques like super-resolution may transition from necessity to option. Understanding when such preprocessing adds value and when it is redundant with architectural robustness is important for efficient pipeline design in the next generation of STR systems.

6 References

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